



DEGENERATIVE DISORDERS



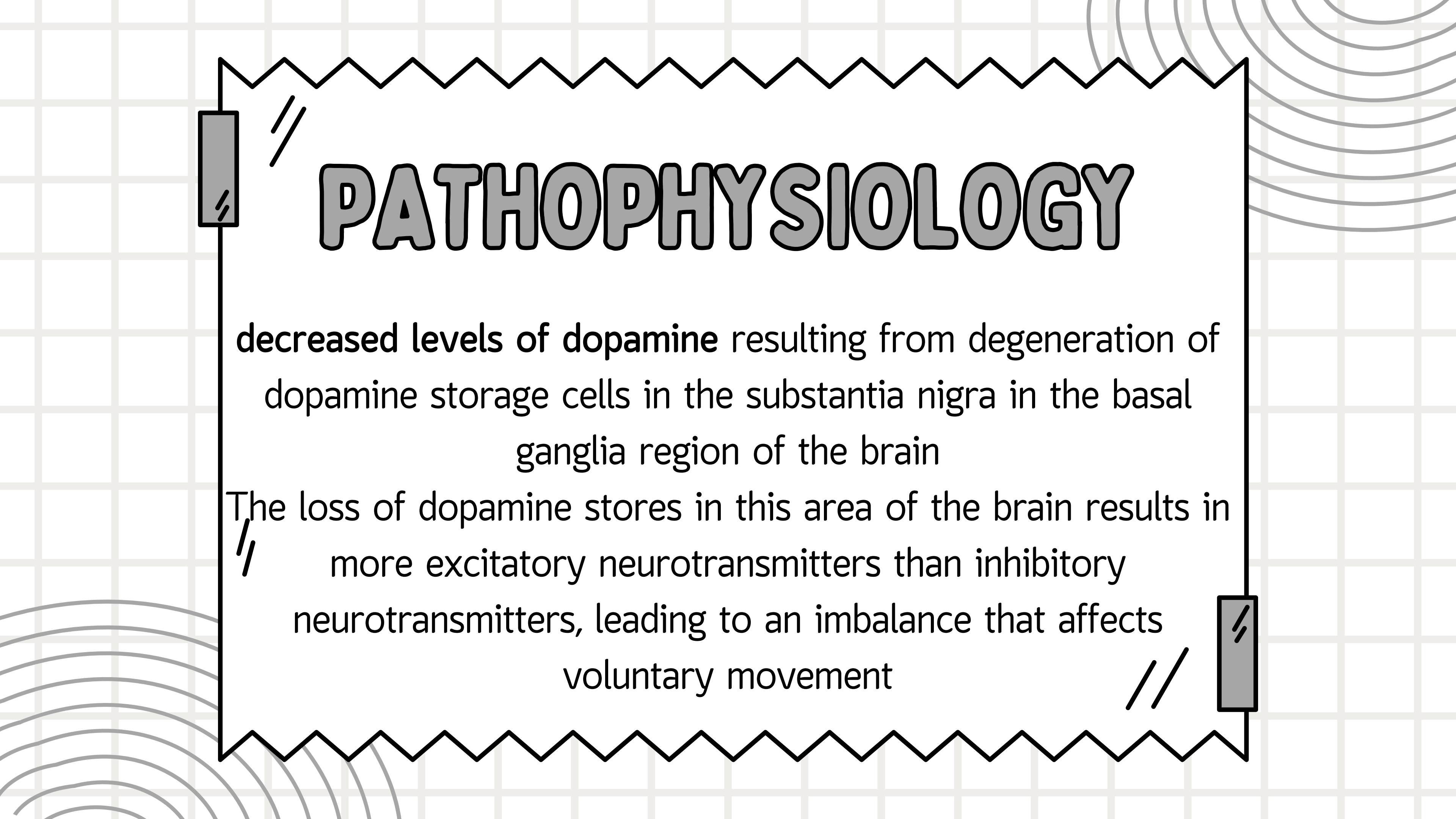


PARKINSON'S DISEASE

WHAT IS PARKINSON'S?

It is a progressive
neurological condition
of the brain.



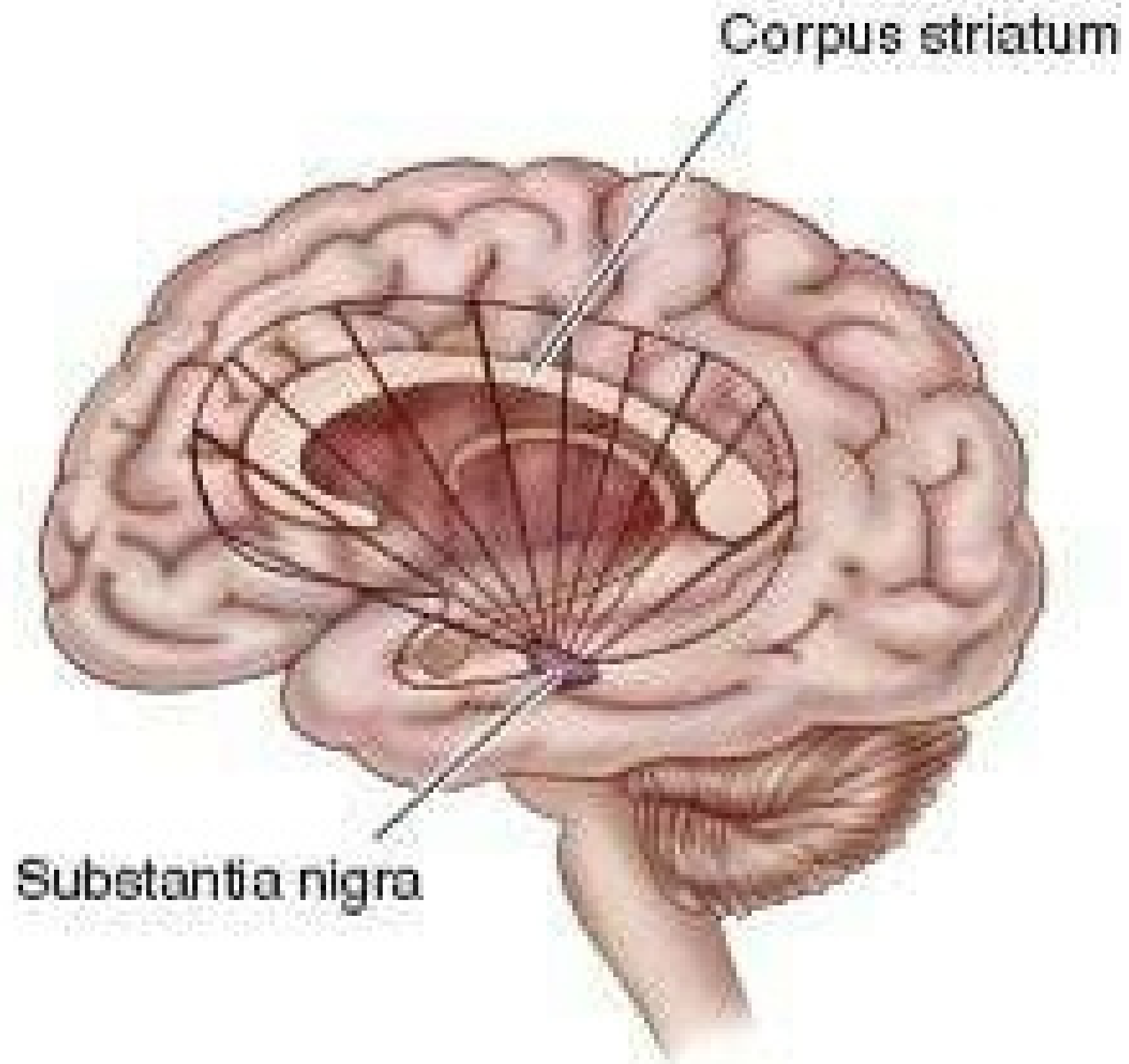


PATHOPHYSIOLOGY

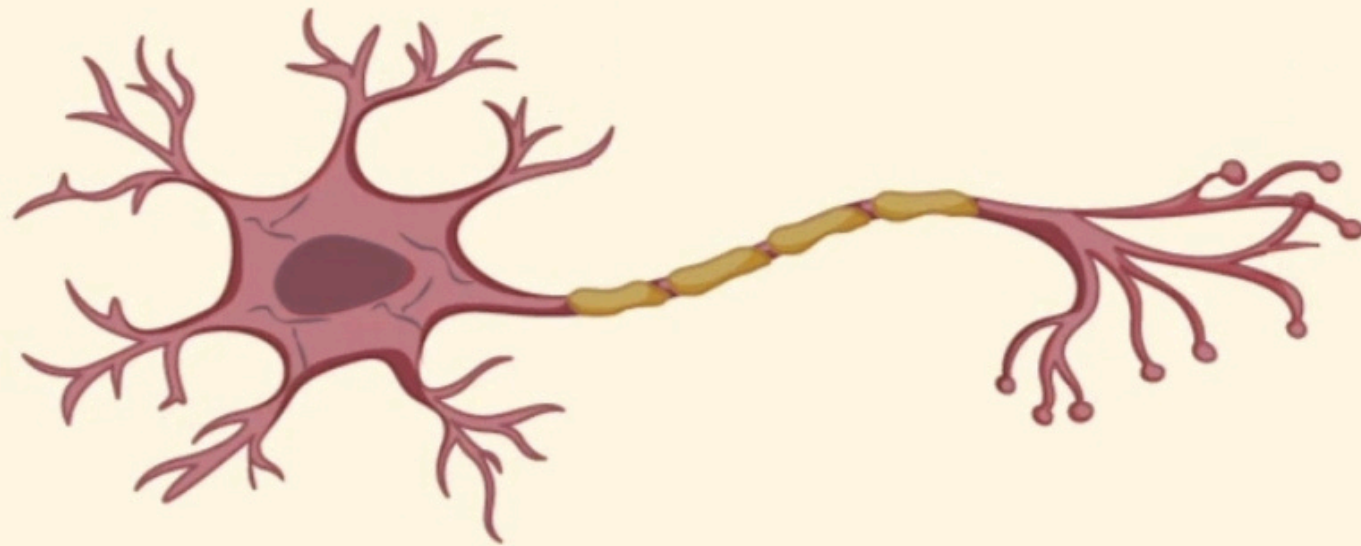
decreased levels of dopamine resulting from degeneration of dopamine storage cells in the substantia nigra in the basal ganglia region of the brain

The loss of dopamine stores in this area of the brain results in more excitatory neurotransmitters than inhibitory neurotransmitters, leading to an imbalance that affects voluntary movement

Physiology Pathophysiology



PARKINSON'S DISEASE



LOW Dopamine

HIGH Acetylcholine

SYMPTOMS

Memory
problems

Tremor

Slow
movement

Reduced
balance

Rigidity

Depression

PARKINSON'S DISEASE

Signs & Symptoms

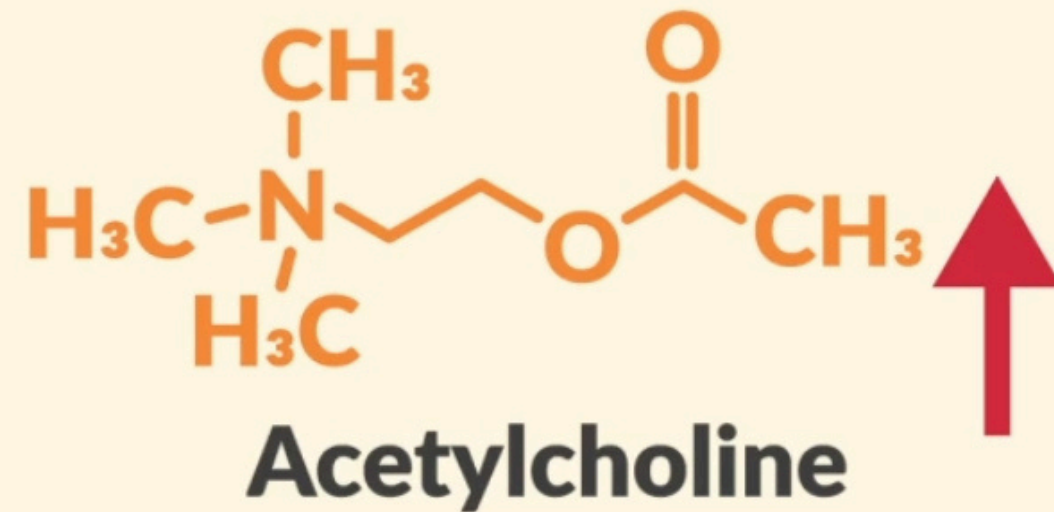


3 Key signs

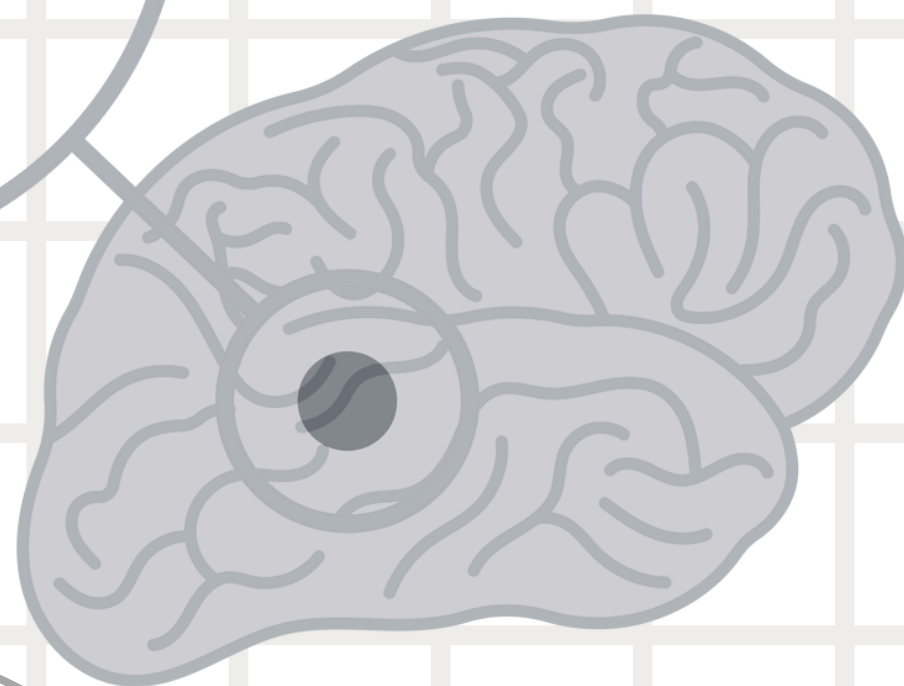
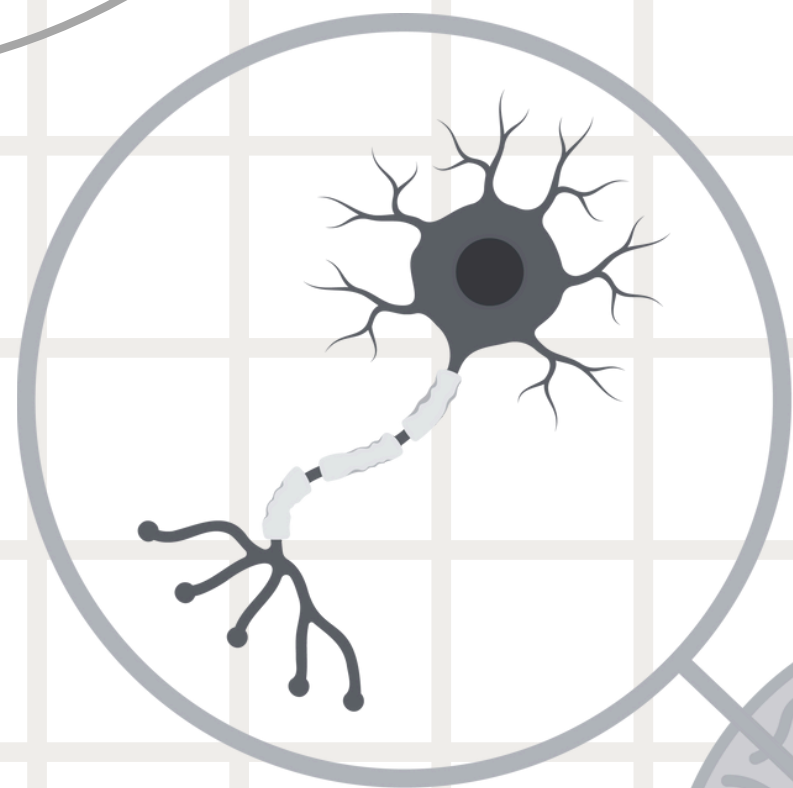
1. Shuffling gait & Decreased arm swinging
2. Pill rolling
3. Tremors at rest

PARKINSON'S DISEASE

Signs & Symptoms

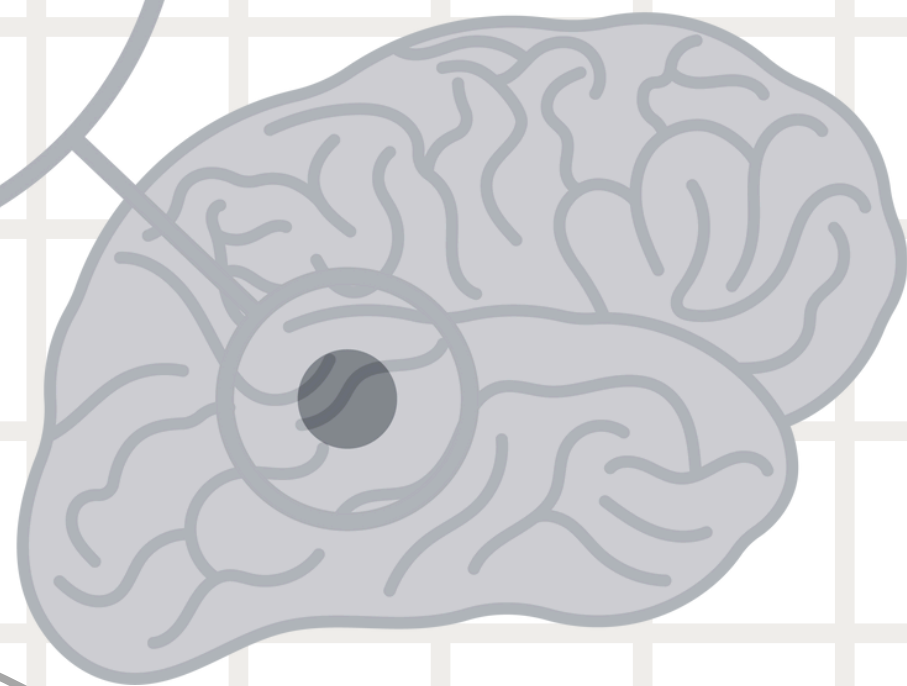
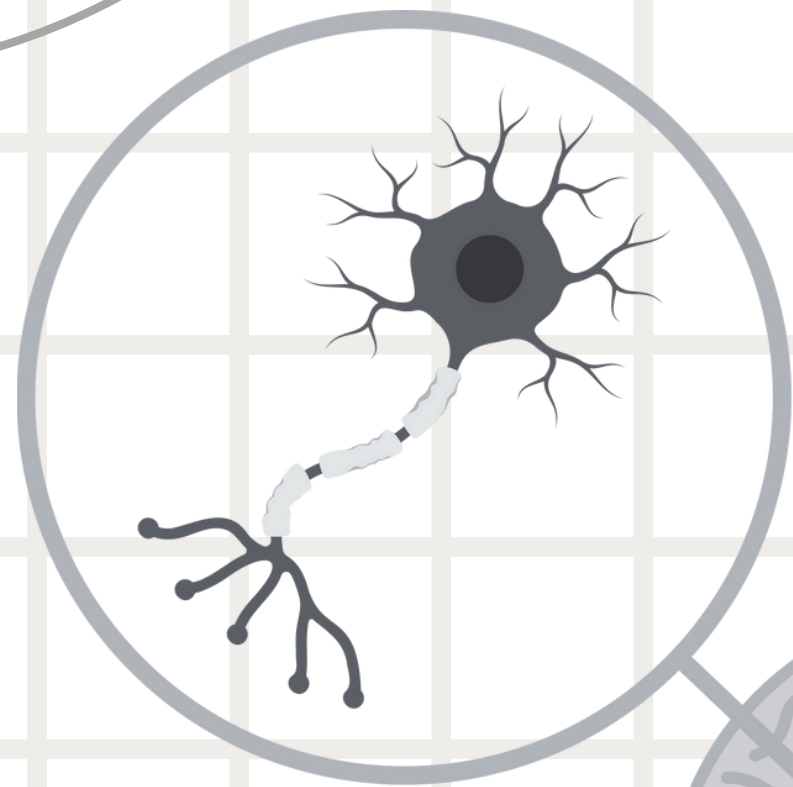


1. Dysphagia
2. Drooling



CAUSES

Nerve cells within the substantia nigra, which is part of the basal ganglia structure of the brain, are reduced. This reduces the chemical dopamine, producing the symptoms of Parkinson's.



CAUSES

causative factors, including genetics, atherosclerosis, excessive accumulation of oxygen free radicals, viral infections, brain trauma, chronic use of antipsychotic medications, and some environmental exposures

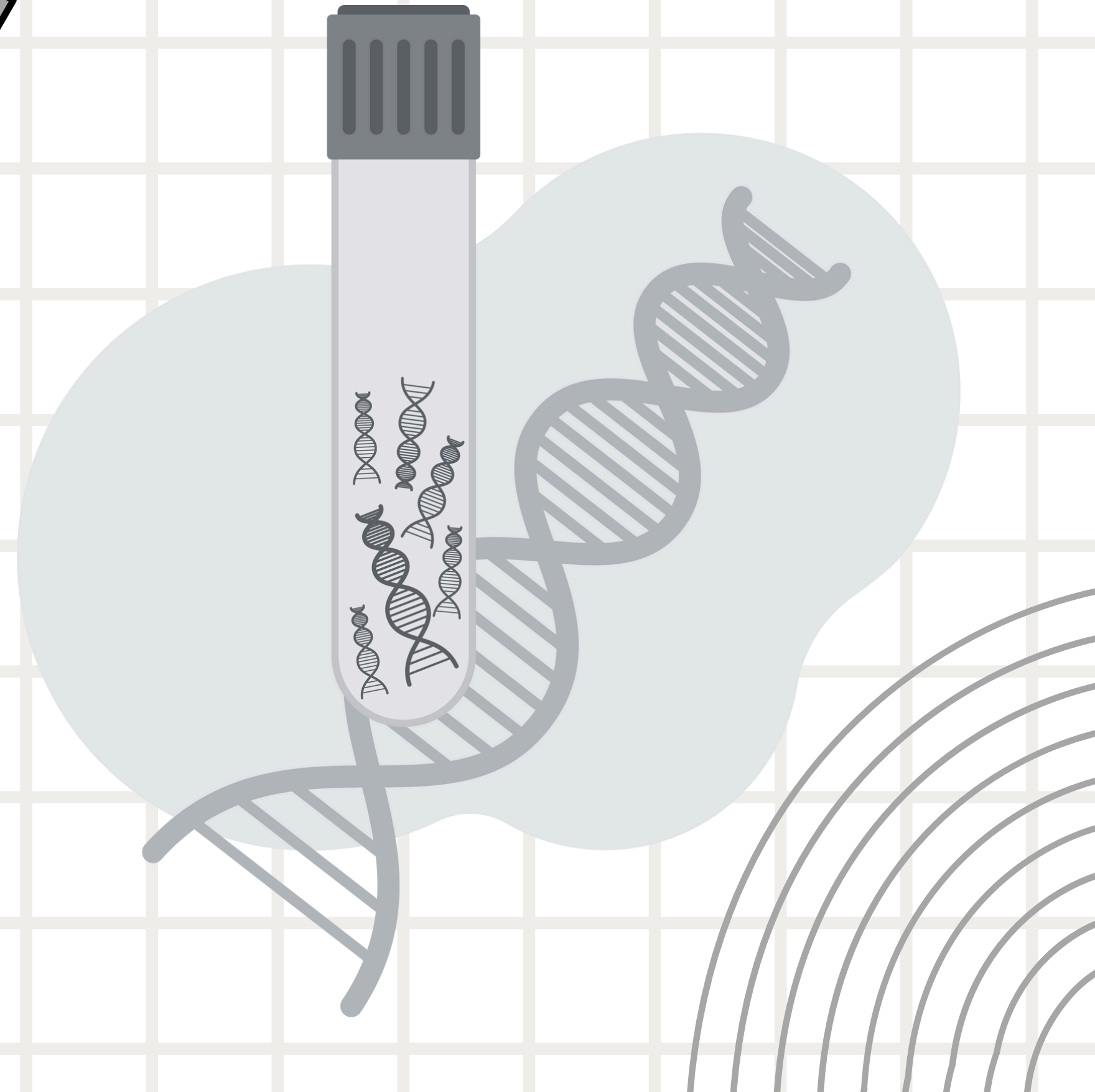
RISK FACTORS

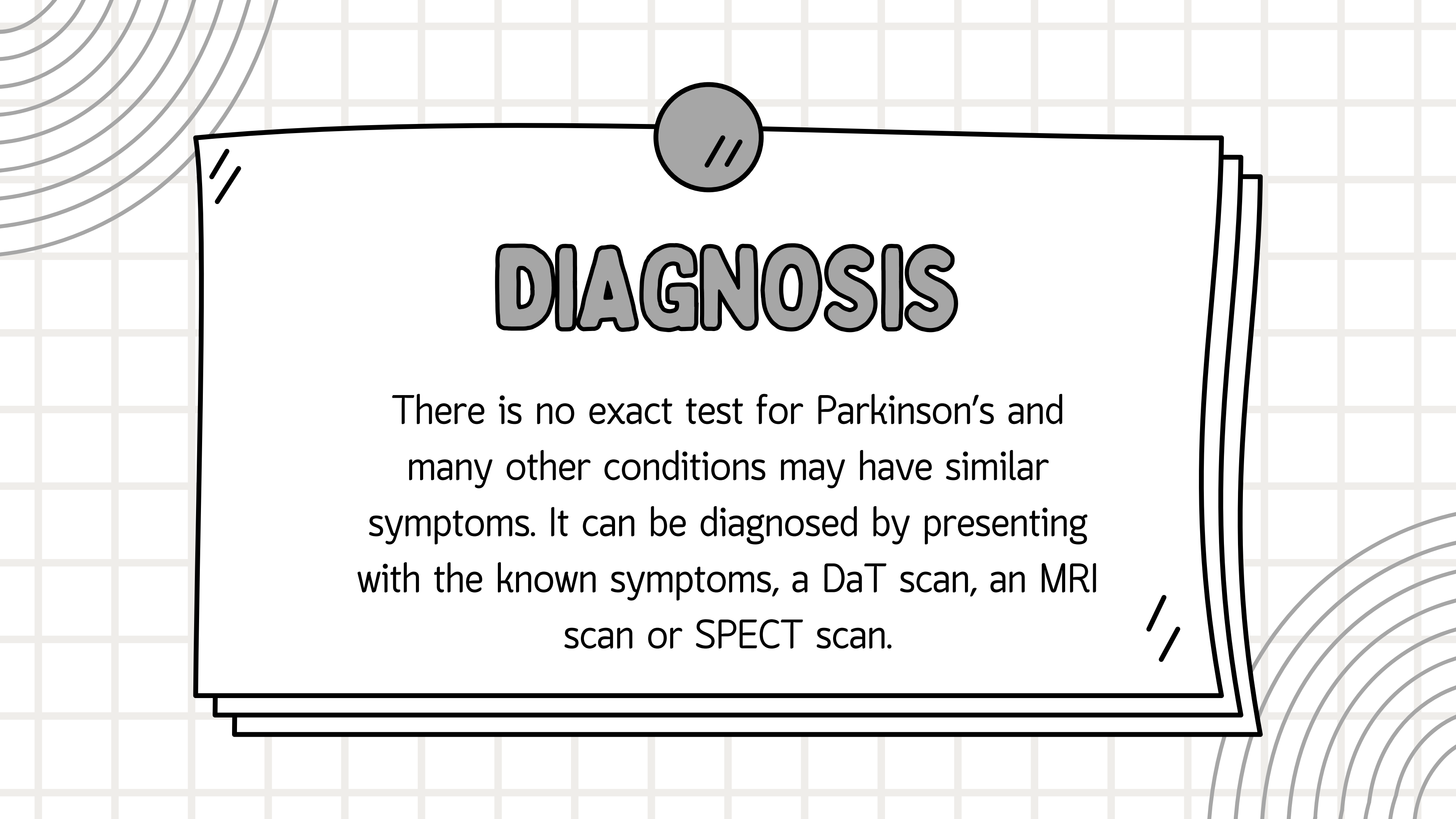
Pesticide
exposure

Genetics

Being Male

Being over
50





DIAGNOSIS

There is no exact test for Parkinson's and many other conditions may have similar symptoms. It can be diagnosed by presenting with the known symptoms, a DaT scan, an MRI scan or SPECT scan.

PARKINSON'S DISEASE

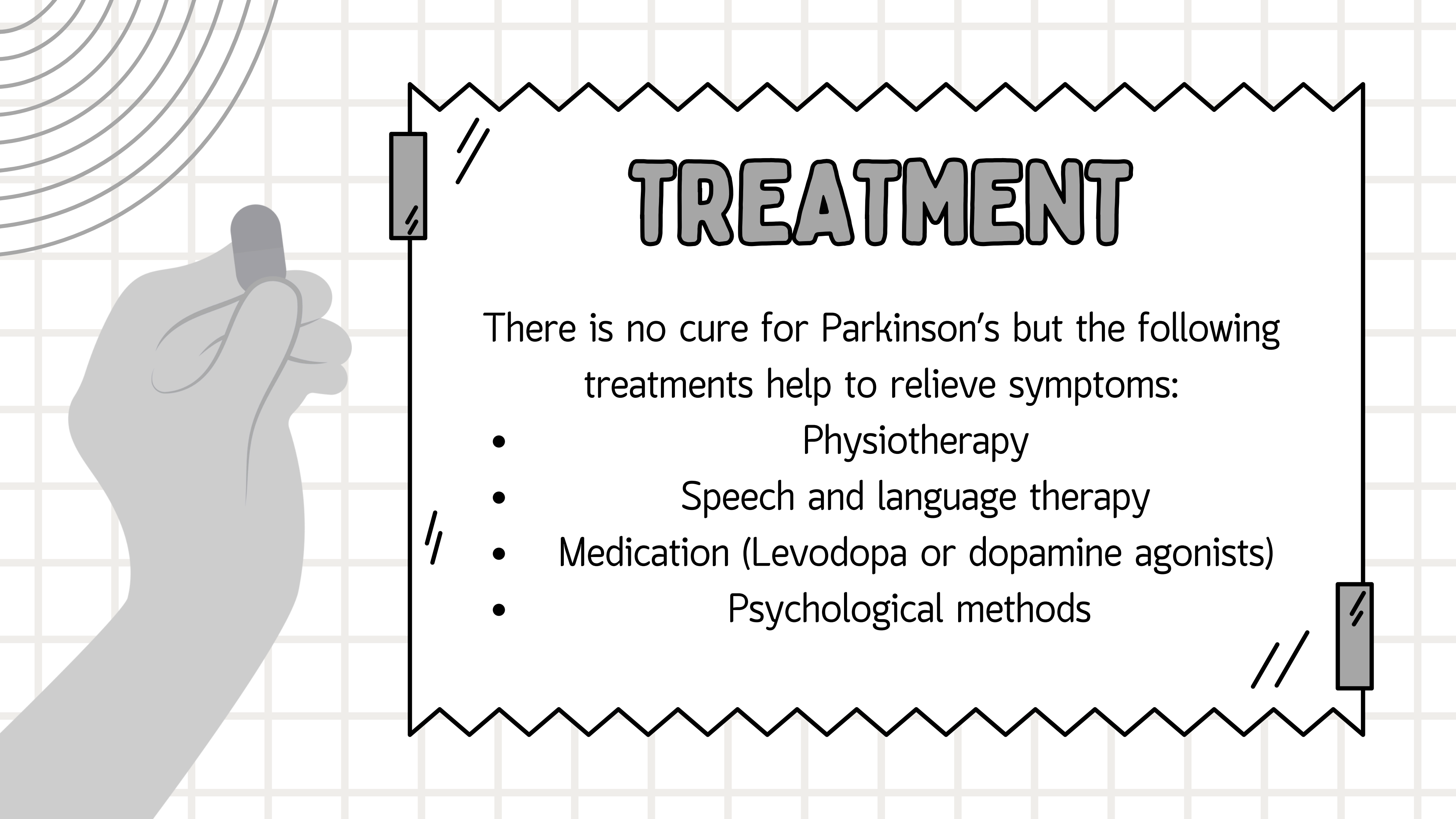
Diagnostics



HESI

Diagnostic tests for Parkinson's?
Select All That Apply

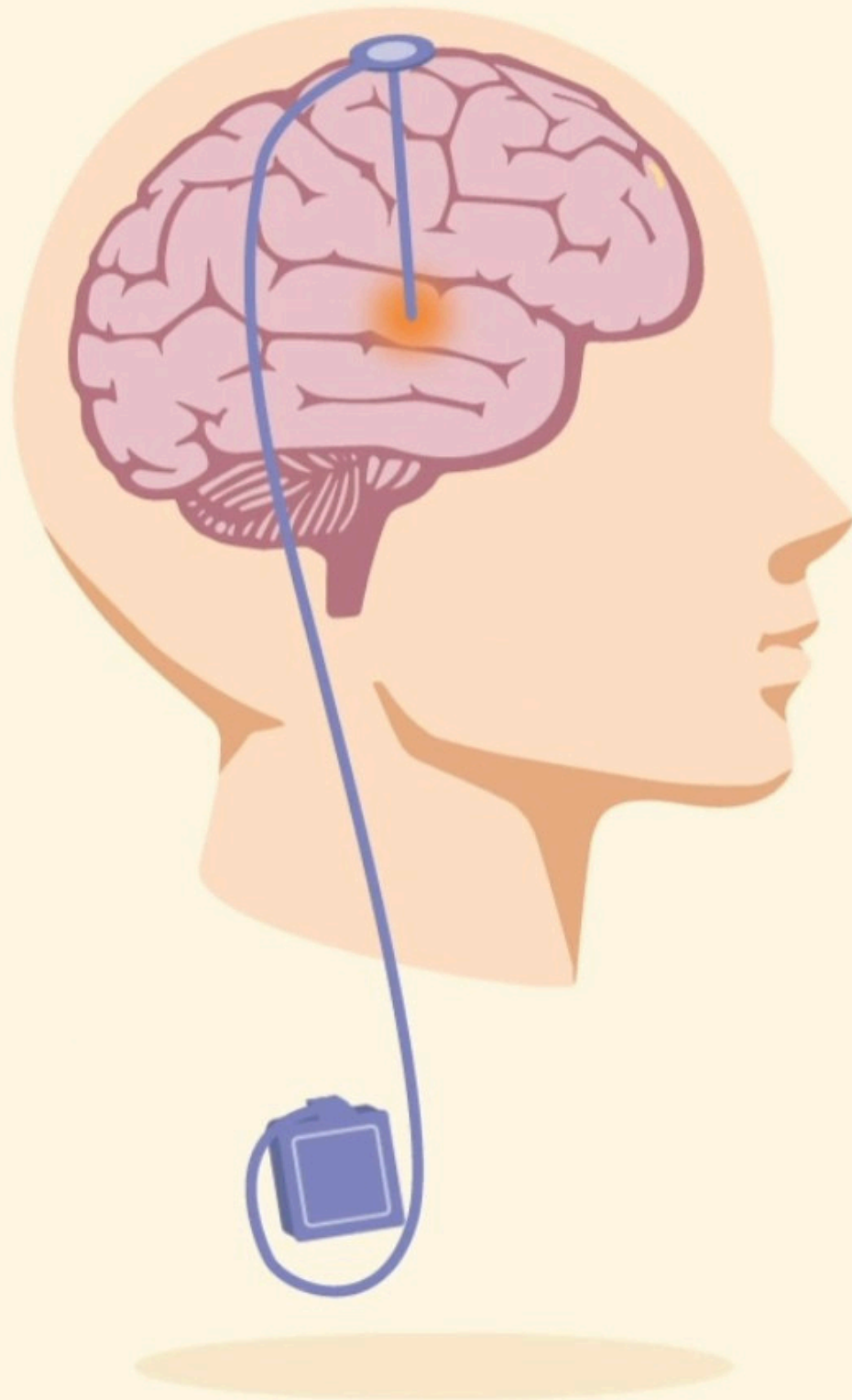
- ✓ ☒ CT brain - cerebral atrophy wasting away or destruction of brain tissue
- ✓ ☒ Decreased motility in pt. Upper GI tract
- ✓ ☒ Positive response to low dose carbidopa levodopa



TREATMENT

There is no cure for Parkinson's but the following treatments help to relieve symptoms:

- Physiotherapy
- Speech and language therapy
- Medication (Levodopa or dopamine agonists)
- Psychological methods



Surgical Options for Parkinson

- Ablation
- Transplantation
- Deep brain stimulation

PARKINSON'S DISEASE

Critical complications & Nursing Interventions



Airway = #1

- Suction at bedside (excess drooling)
- **Eating**
 - Pureed Diet or "Small Bite sized pieces"
 - Tissues readily **available during eating**
 - "Add **thickening** agent to **fluids**"
 - **Dysphagia** - Monitor swallowing
 - **HOB up** - High Fowler's "Sit Upright"
- Key words
 - Thicken FLUIDS & Head of Bed UP or Sit UP

PARKINSON'S DISEASE

Critical complications & Nursing Interventions



How to **maintain safety** & also prevent constipation and malnutrition ... **Parkinson's?**
Select All That Apply

- ✓ ☒ Cut food into **bite sized** pieces
- ✓ ☒ Make **tissues readily available** during eating
- ✓ ☒ Include **whole grains** and **fruits in diet**

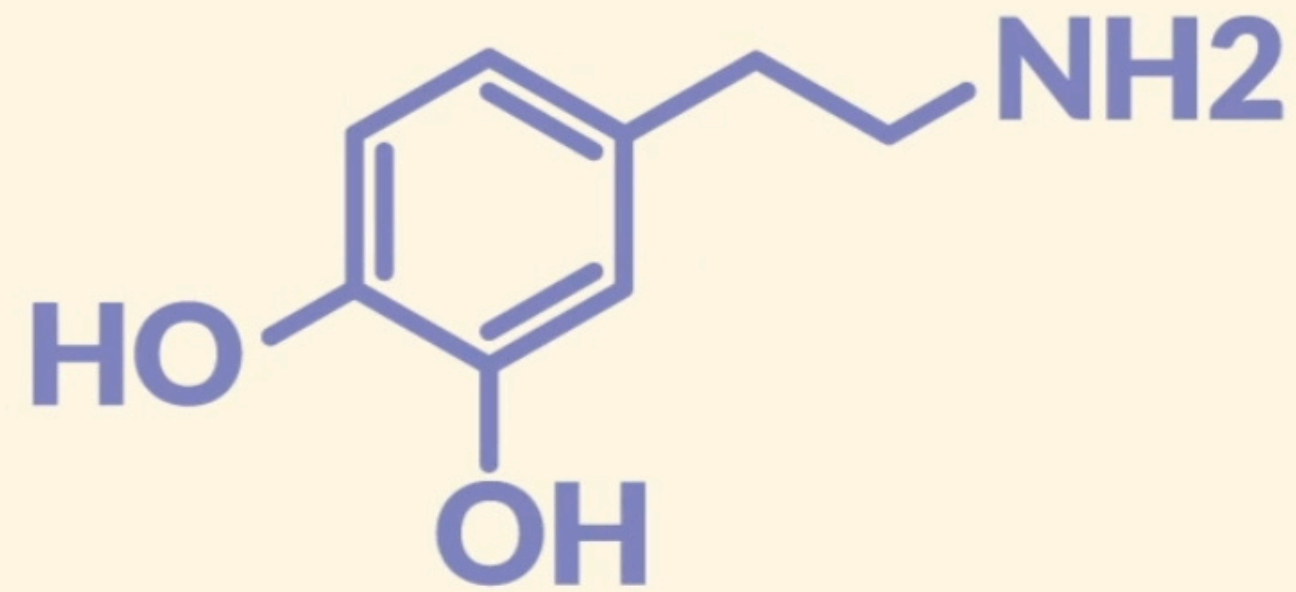


Realistic goal...
Parkinson's disease?

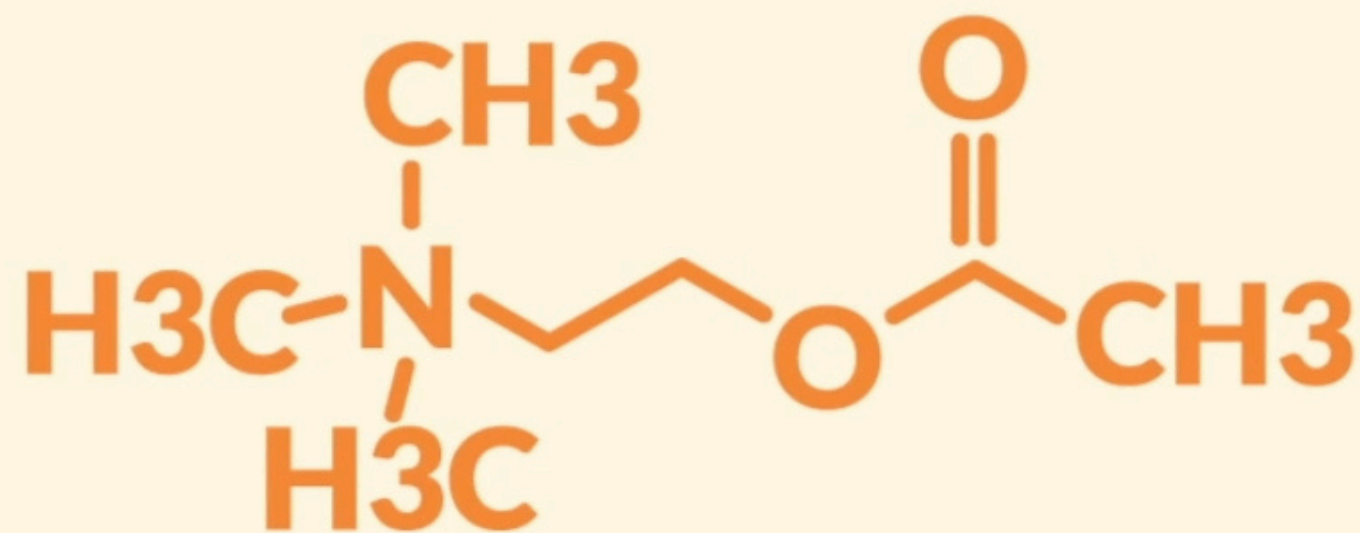
Maintain optimal function
within the client's limitations.

PARKINSON'S DISEASE

Pharmacology



Dopamine

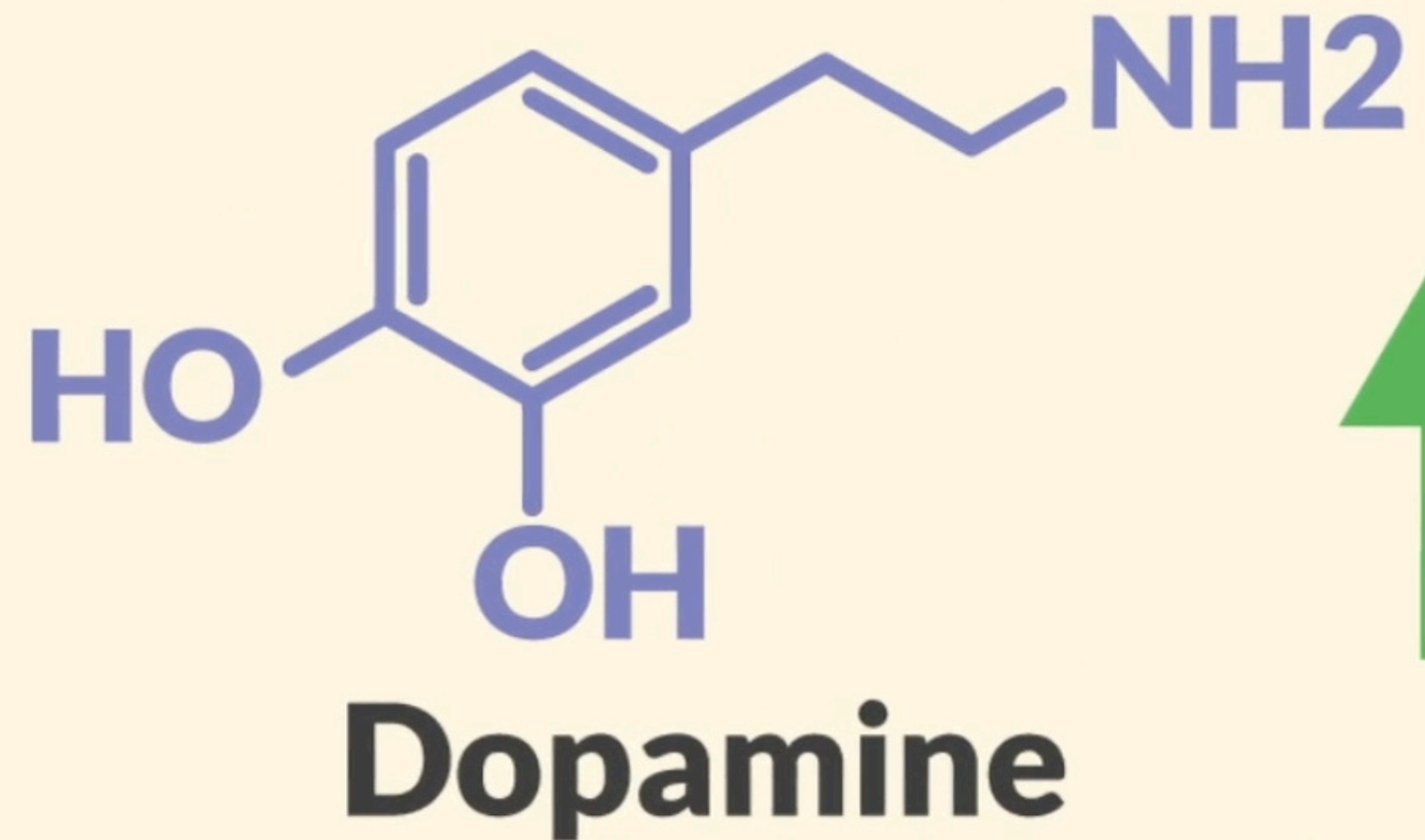


Acetylcholine



PARKINSON'S DISEASE

Pharmacology



PARKINSON'S DISEASE

Pharmacology



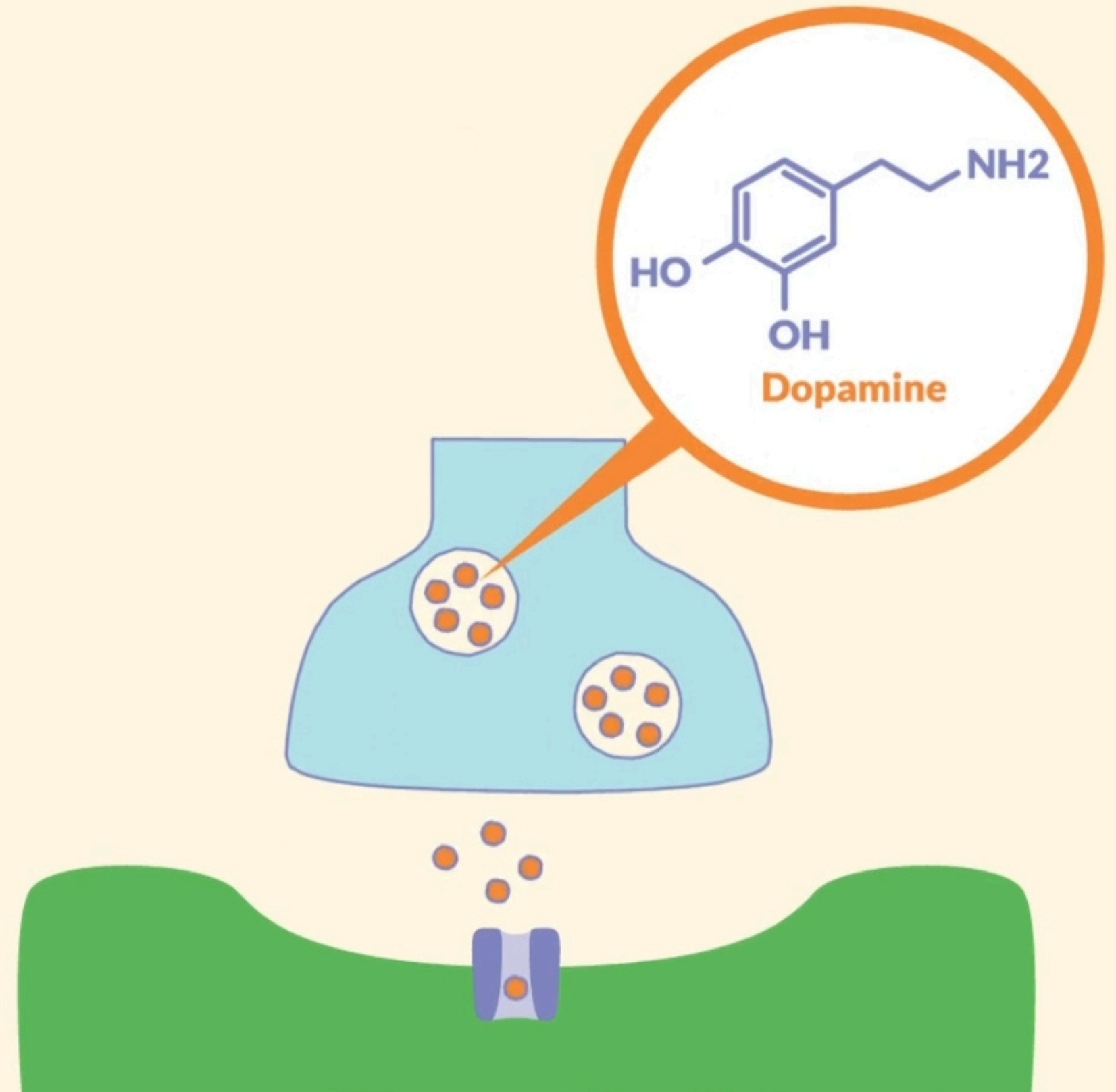
avoid high protein diet

Taking Levodopa Statement
reflecting need for further education:

**“I will buy lots of bone broth
and soup for my parent”**

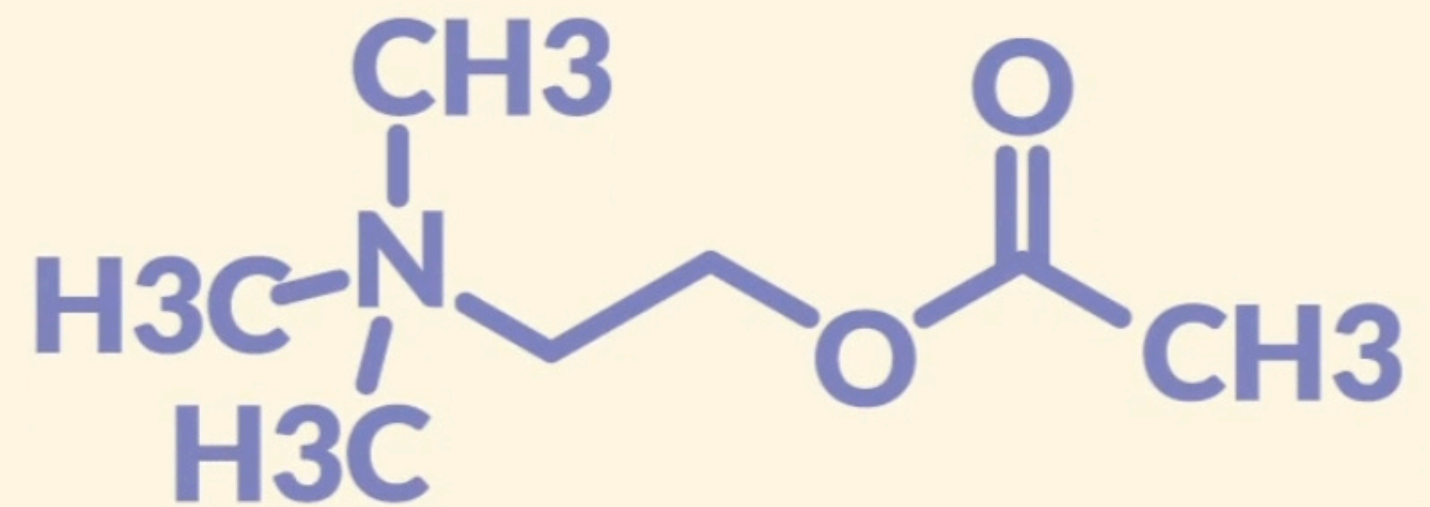
PARKINSON'S DISEASE

Pharmacology



PARKINSON'S DISEASE

Pharmacology



Acetylcholine



Tropine

Tremors



PARKINSON'S DISEASE

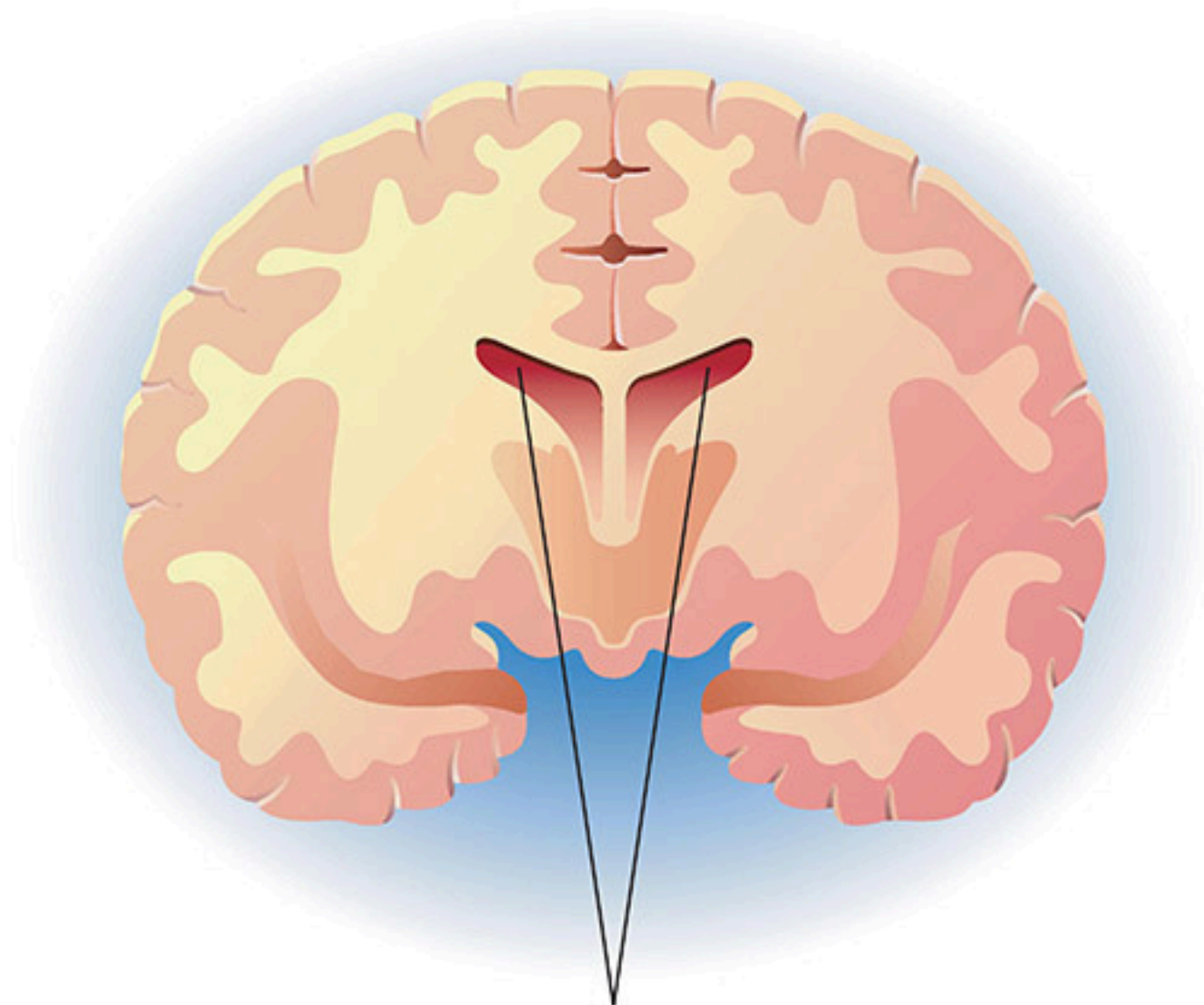
Pharmacology

MAOIs



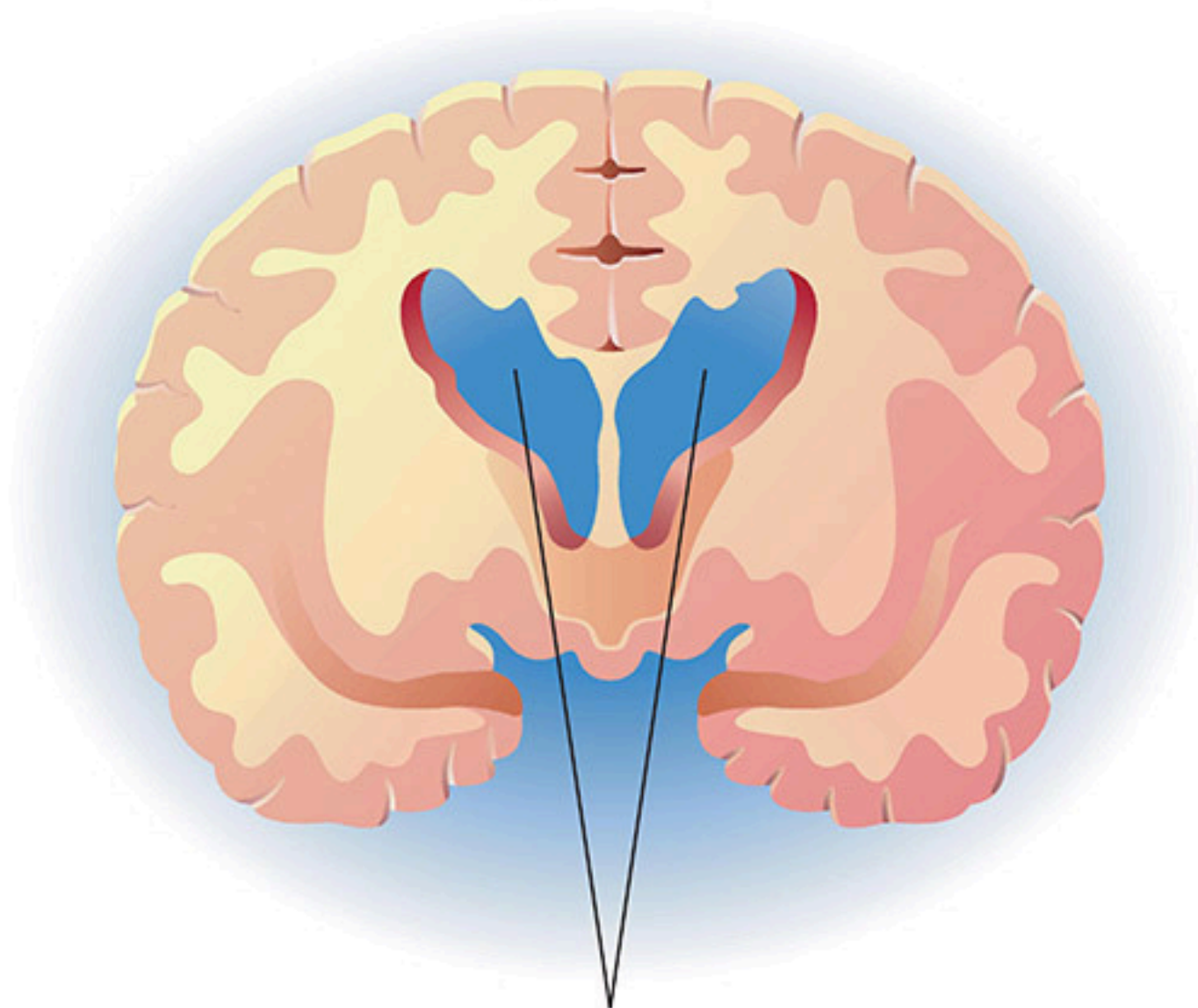
Huntington Disease

Normal brain section



Normal frontal horns
of the lateral ventricles

Huntington's disease

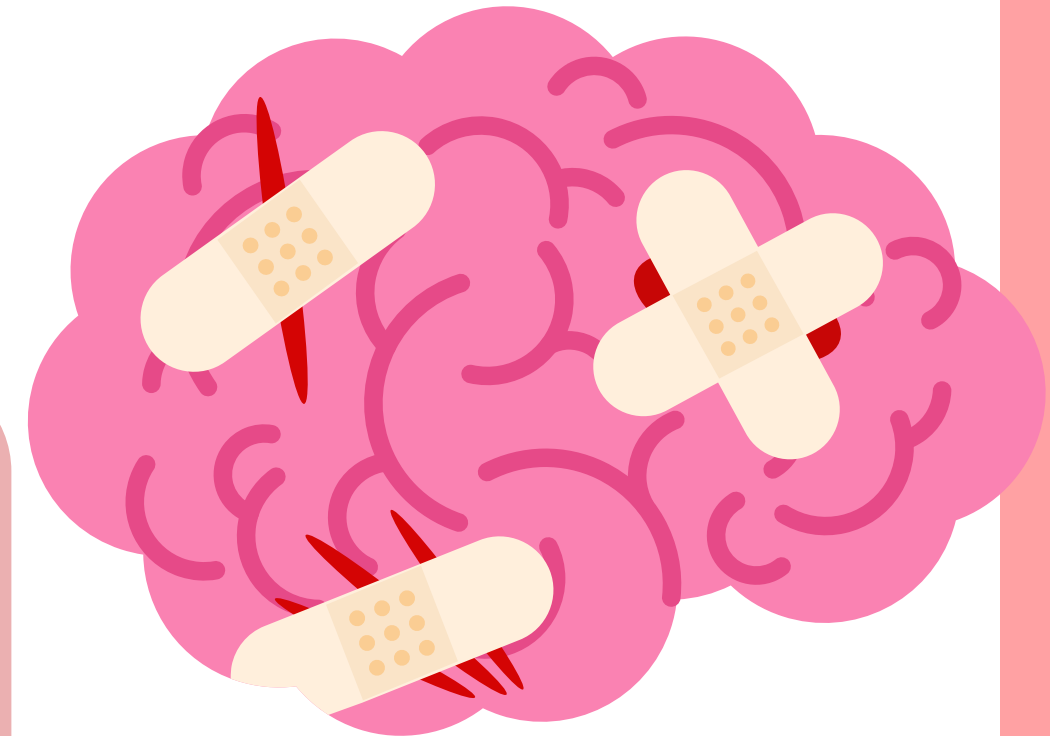


Enlargement of the frontal horns
of the lateral ventricles

what is huntington diseases?

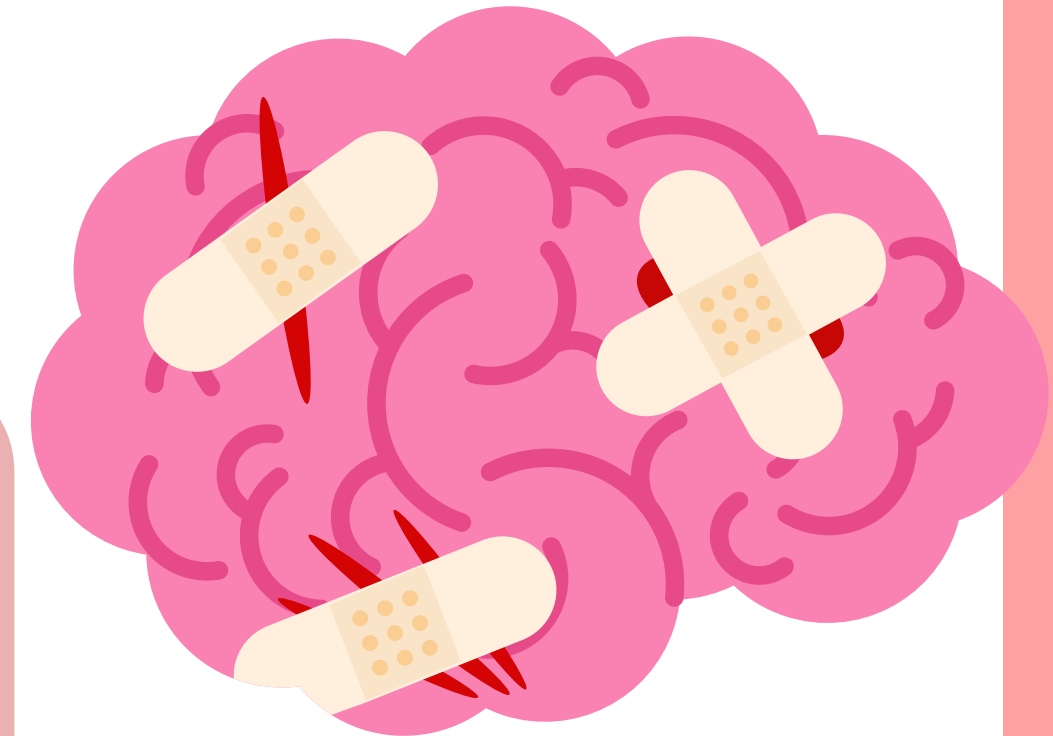
a chronic, progressive, hereditary disease of the nervous system that results in progressive involuntary choreiform movement and dementia

It is transmitted as an autosomal dominant genetic disorder; therefore, each child of a parent with Huntington disease has a 50% risk of inheriting the disorder



Pathophysiology

premature death of cells in the striatum (caudate and putamen) of the basal ganglia, the region deep within the brain that is involved in the control of movement. Cells also are lost in the cortex, the region of the brain associated with thinking, memory, perception, judgment, and behavior, and in the cerebellum, the area that coordinates voluntary muscle activity



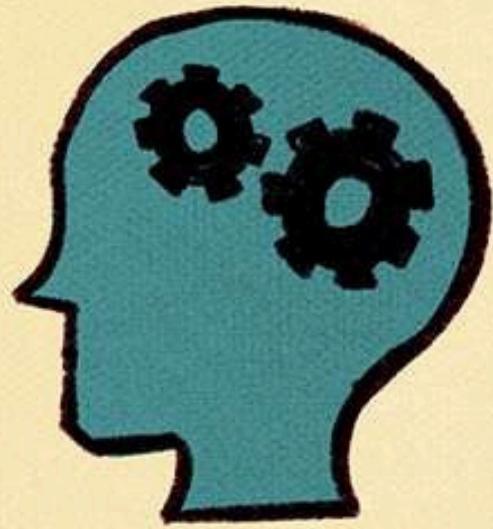
HUNTINGTON DISEASE



Key Point

- Passed on by Genetics
- Only requires 1 parent!
- Recommend Genetic counseling

Common Symptoms of Huntington's Disease



Cognitive decline



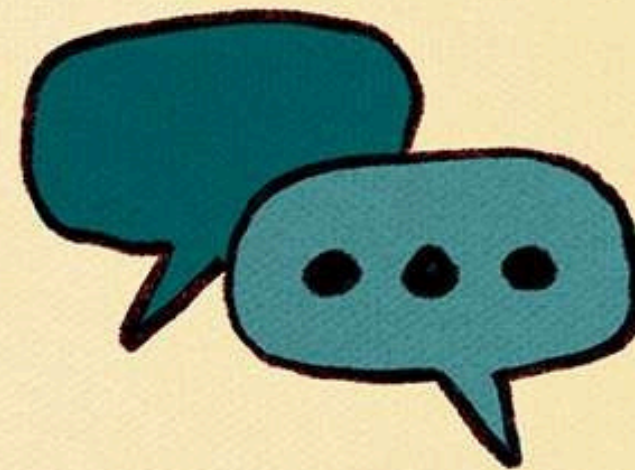
Mood swings



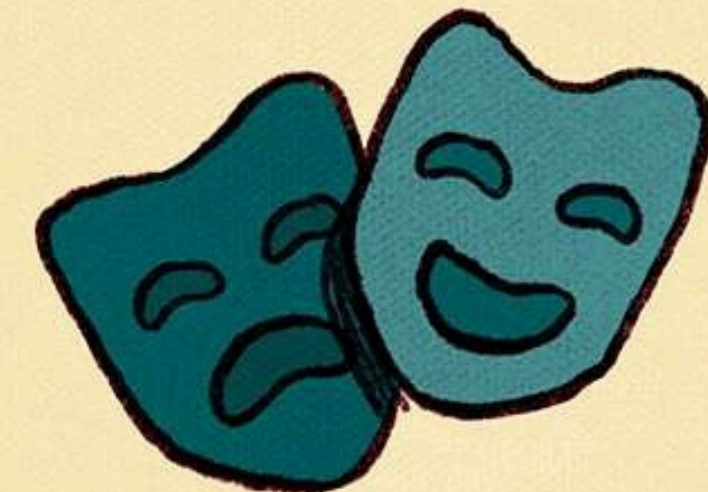
CHOREA
(RAPID, JERKY, INVOLUNTARY,
PURPOSELESS MOVEMENTS)



Hallucinations



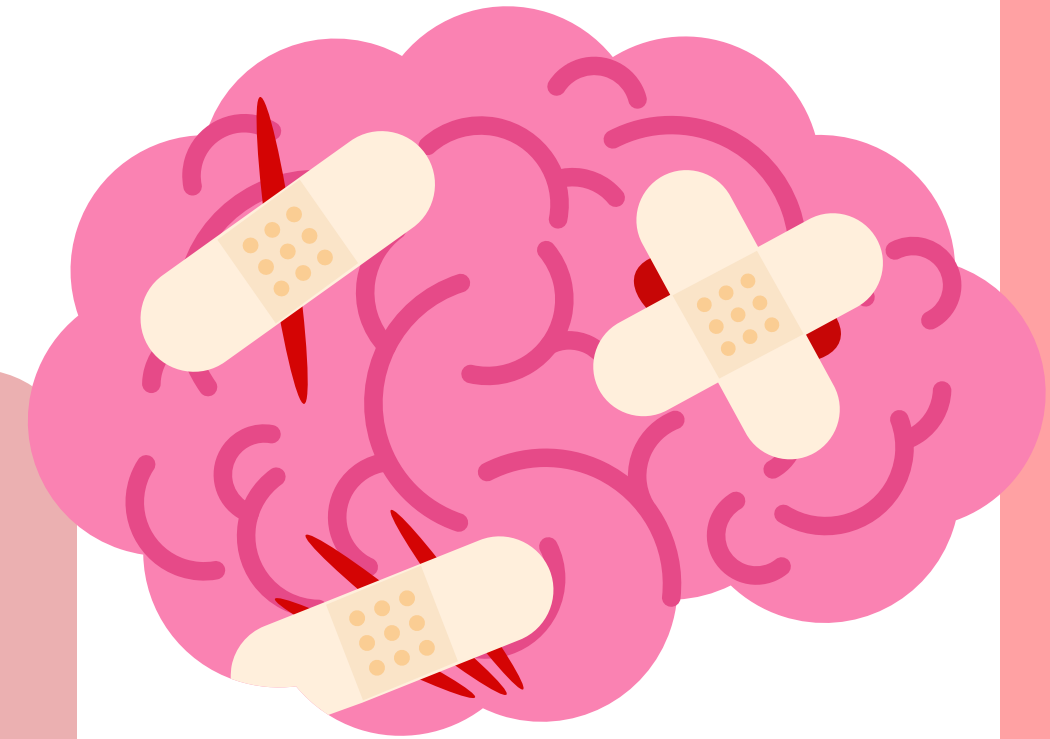
Speech difficulties



Behavioral &
personality changes

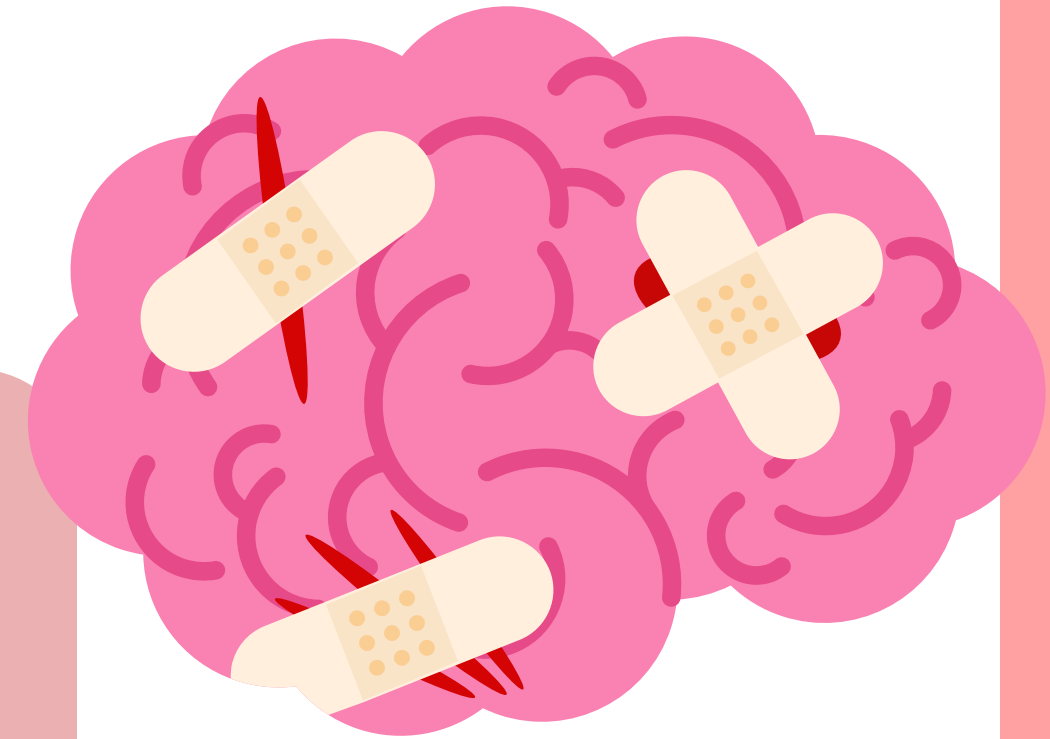
Assessment and Diagnostic Findings

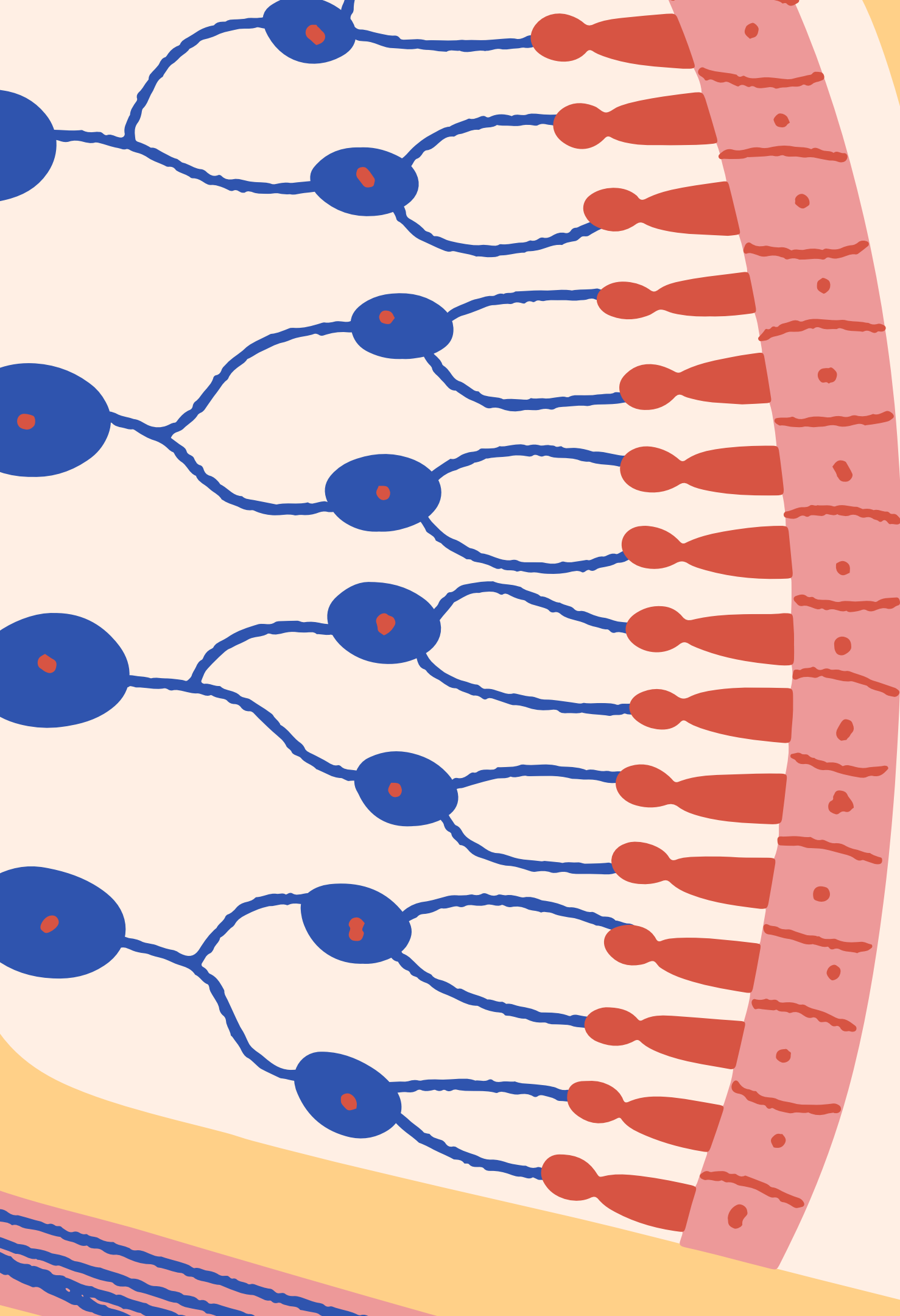
diagnosis is made based on the clinical presentation of characteristic symptoms, a positive family history, the known presence of a genetic marker, and exclusion of other causes.



Management

- no treatment halts or reverses the underlying process
- **Tetrabenazine** (Xenazine) is the only approved drug for the treatment of the chorea
- **thiothixene hydrochloride** (Navane) and **haloperidol decanoate** (Haldol), which predominantly block dopamine receptors
- Selective serotonin reuptake inhibitors **SSRI** and tricyclic antidepressants **TCA** have been recommended for control of psychiatric symptoms





AMYOTROPHIC LATERAL SCLEROSIS



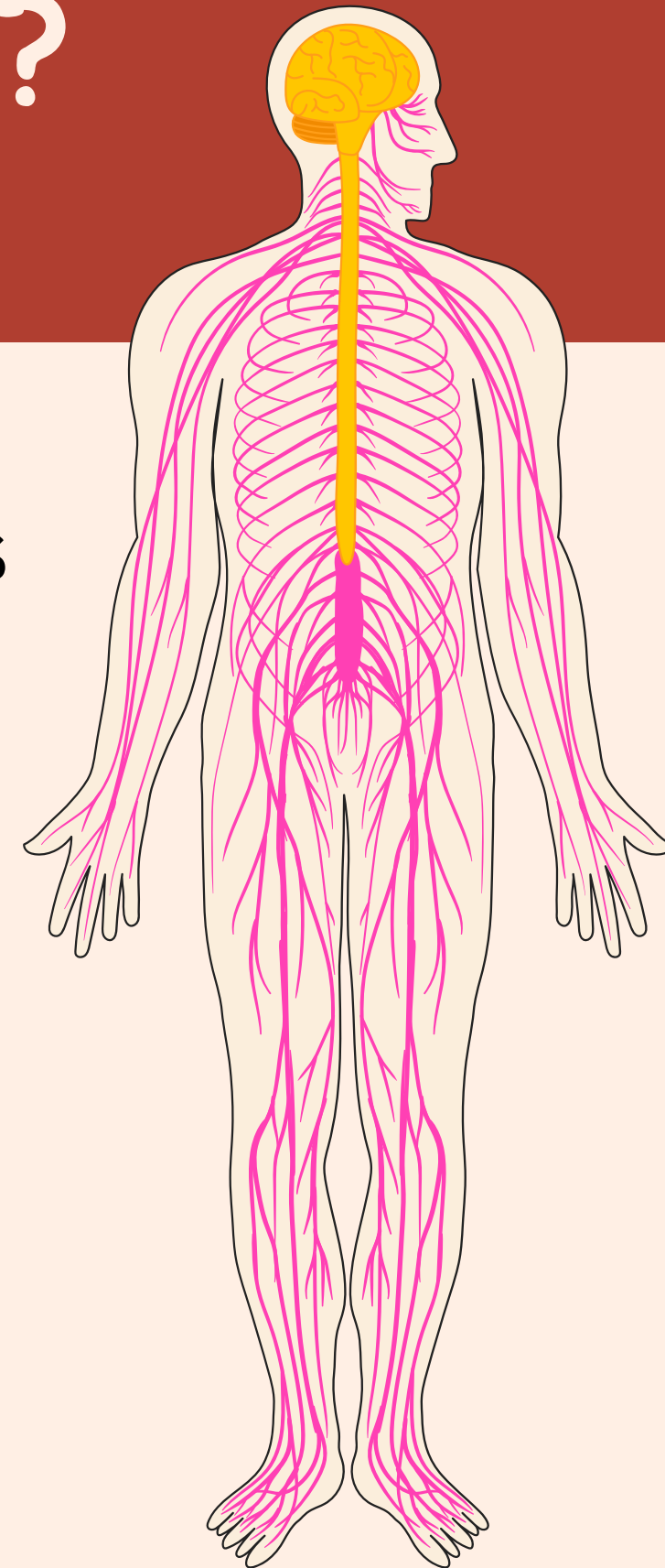
what is Amyotrophic Lateral Sclerosis?

Lou Gehrig's disease

disease of **unknown cause** in which there is a loss of motor neurons (nerve cells controlling muscles) in the anterior horns of the spinal cord and the motor nuclei of the lower brain stem

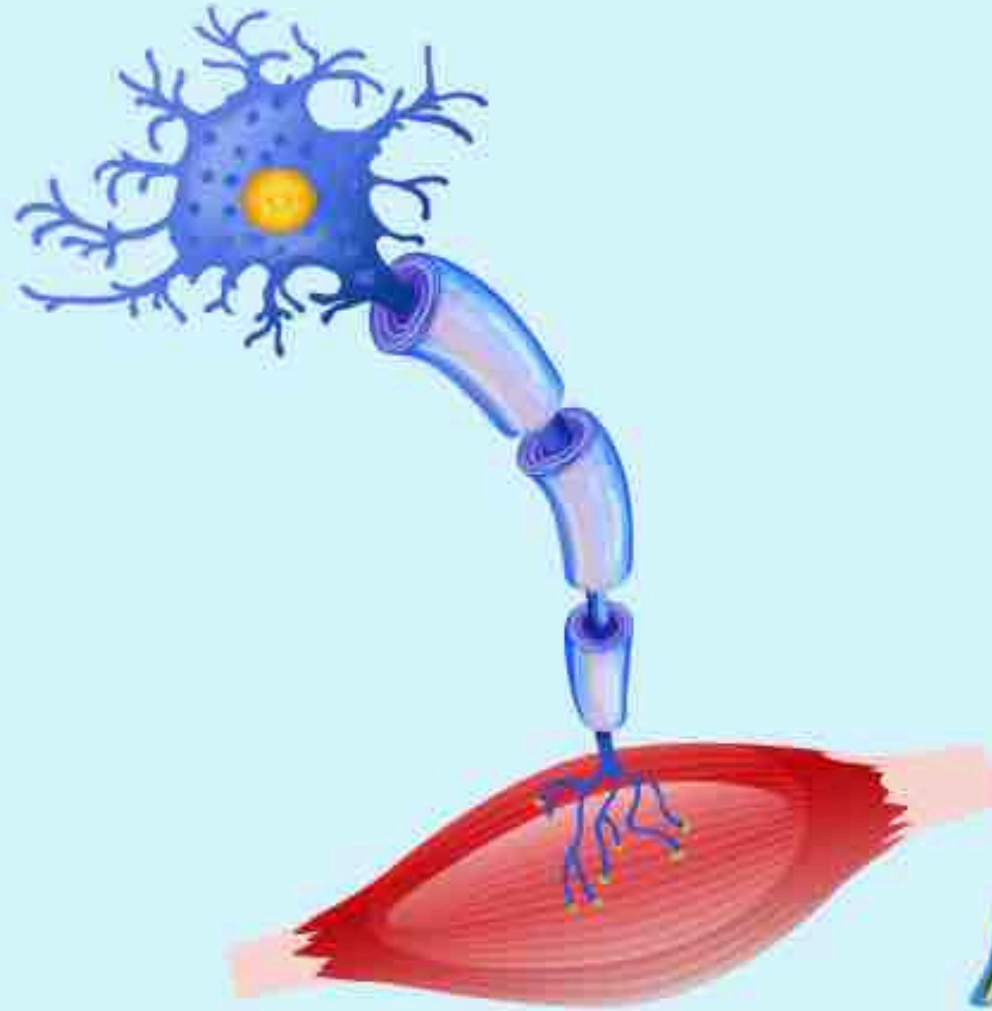
As motor neuron cells die, the muscle fibers that they supply undergo atrophic changes. Neuronal degeneration may occur in both the upper and lower motor neuron systems
overexcitation of nerve cells by the neurotransmitter glutamate results in cell injury and neuronal degeneration.

Possible causes of ALS include **autoimmune disease, free radical damage, and oxidative stress**



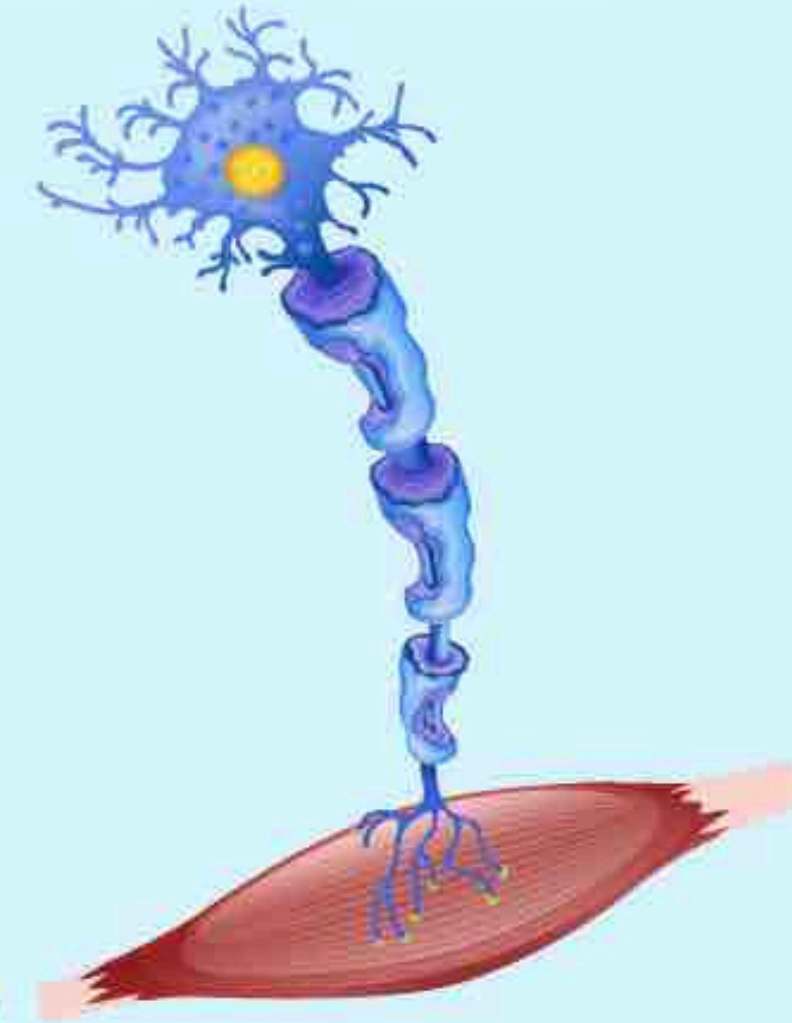
Amyotrophic Lateral Sclerosis (ALS)

normal nerve cell

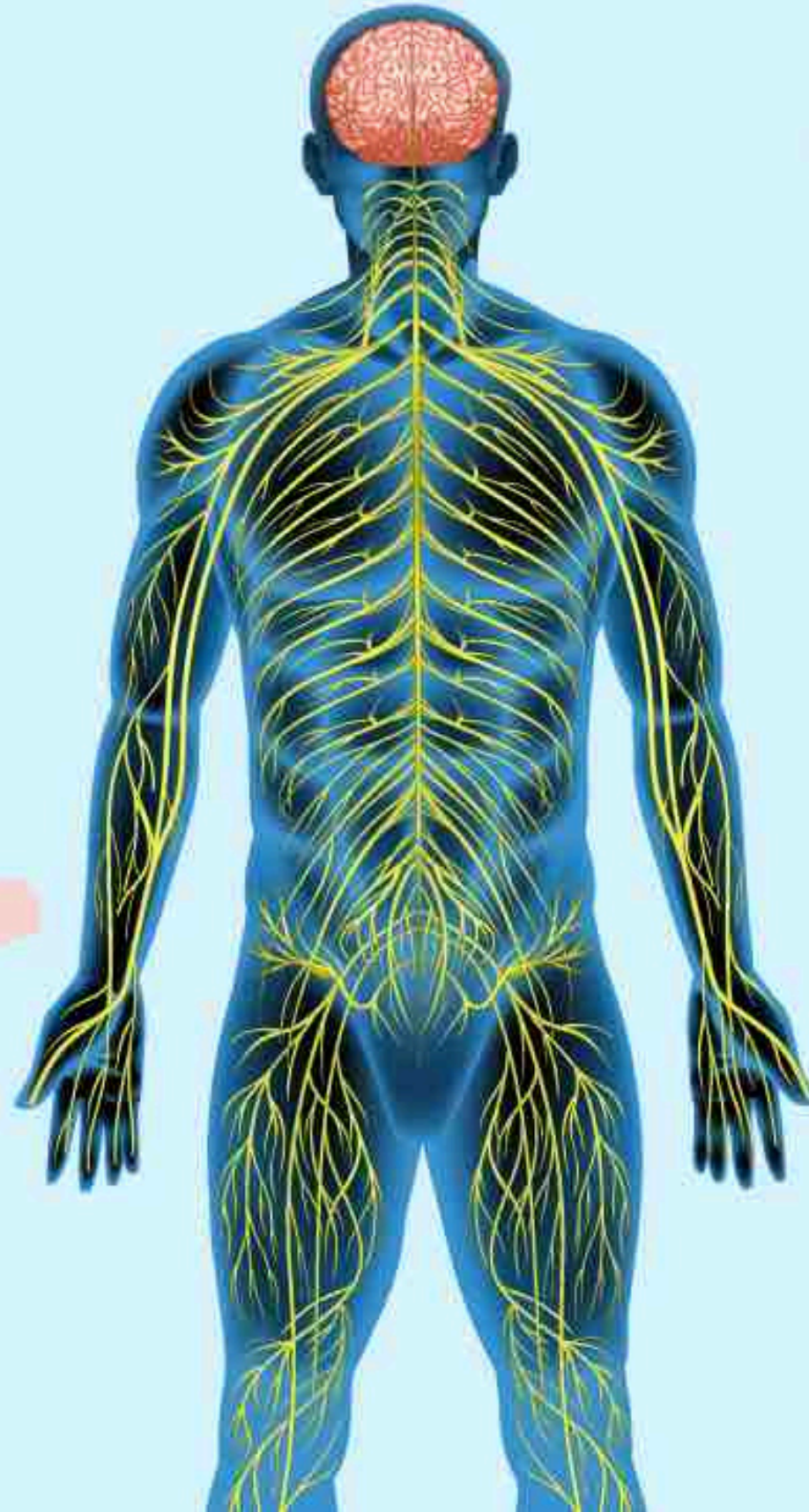


muscle contracts

nerve with sclerosis



muscle unable to contract



ALS STAGES

ALS can be divided into four stages:

Early-stage ALS



- Trouble buttoning clothes or gripping objects
- Balance problems and frequent tripping
- Difficulty swallowing or speech problems
- Muscle twitching and cramping

Middle-stage ALS



- Some muscles become paralyzed
- Walkers and wheelchairs may be needed
- Inability to drive
- Breathing problems

Late-stage ALS



- Mobility is extremely limited
- Most people require a feeding tube
- Inability to communicate without assistance
- Breathing function becomes severely impaired

End-stage ALS



- Paralysis of voluntary muscles
- Respiratory complications are the most common cause of death
- Hospice care may be necessary
- Medications can be given to ease pain and anxiety

A.L.S. AMYOTROPHIC LATERAL SCLEROSIS

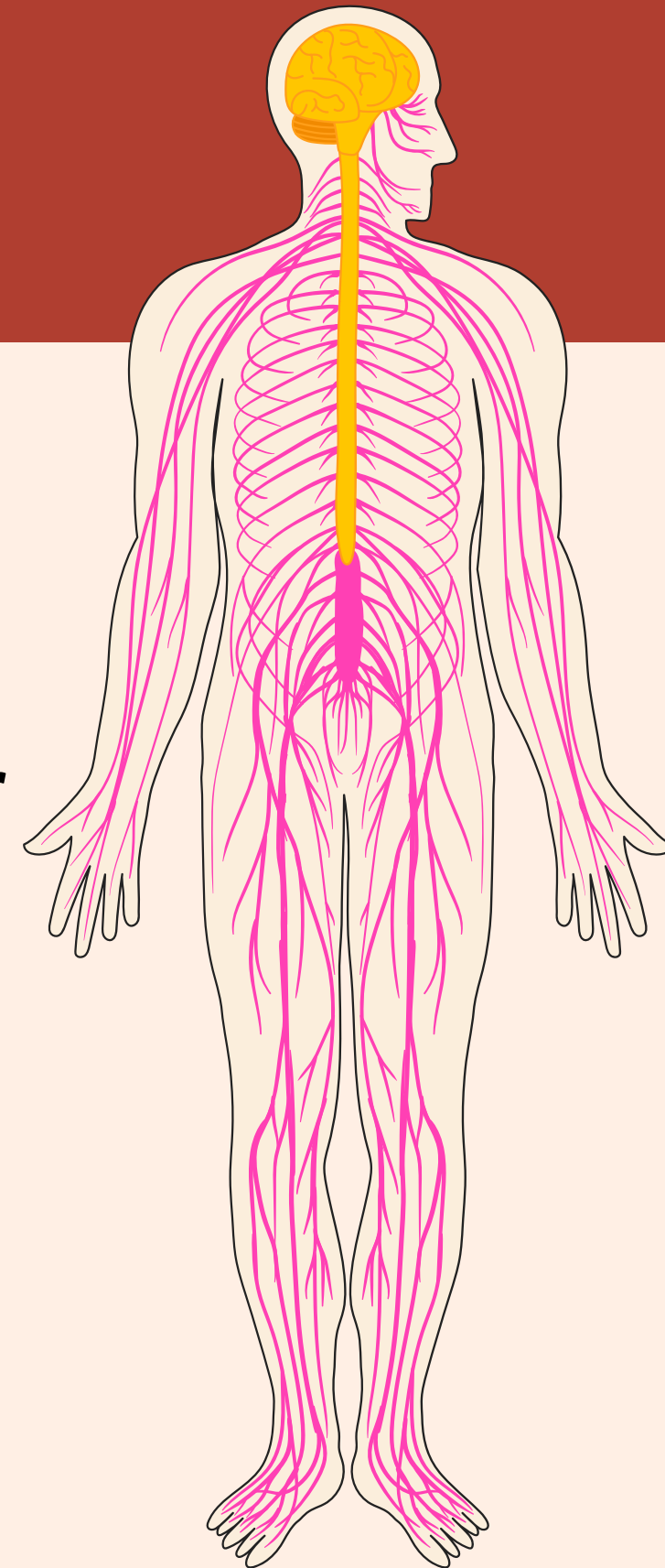
Signs & Symptoms



- Progressive **muscle weakness**
- Dyspnea - **Difficulty breathing**
- Dysphagia - **Difficulty swallowing**
- Dysphasia - Difficulty speaking
- Constipation
- Respiratory failure **#1 Priority**

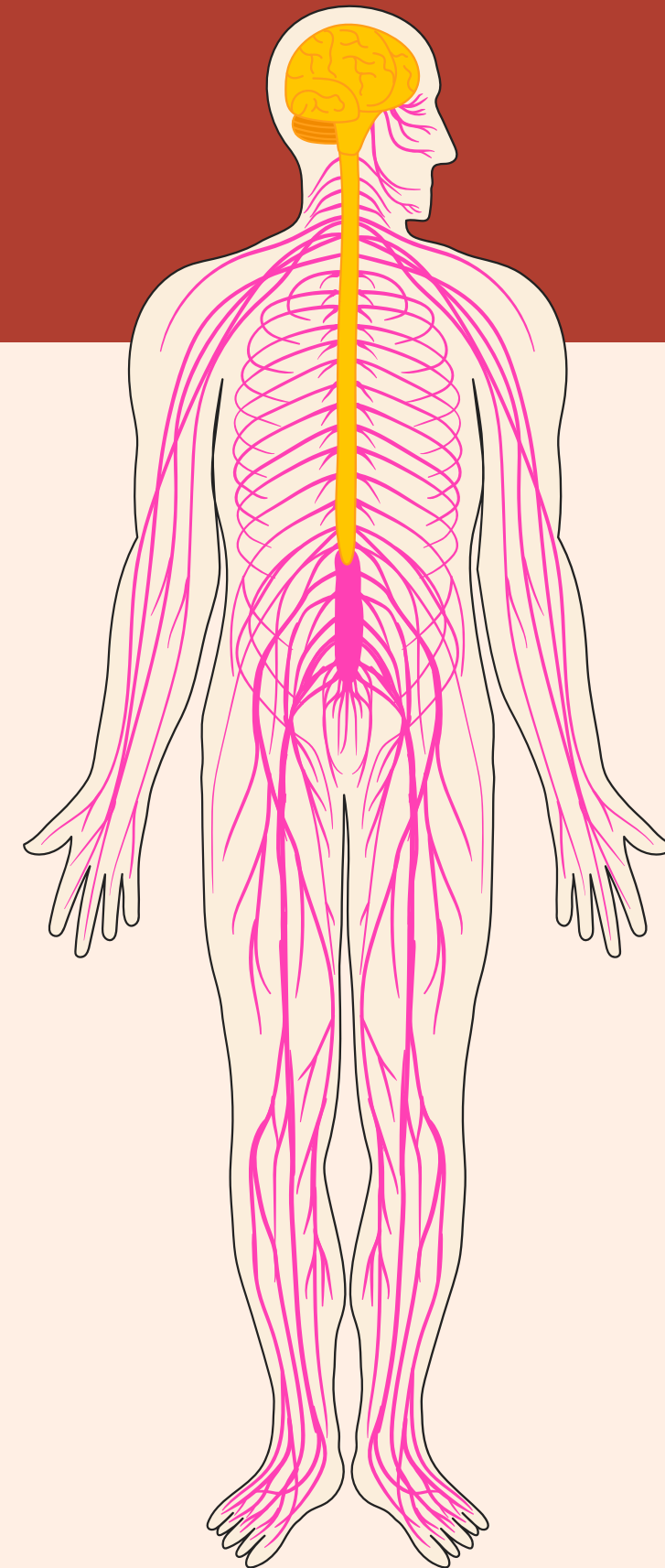
Assessment and Diagnostic Findings

- diagnosed on the basis of the signs and symptoms, because no clinical or laboratory tests are specific to this disease.
- **Electromyography** and **muscle biopsy** studies of the affected muscles indicate reduction in the number of functioning motor units.
- **MRI scan** may show high signal intensity in the corticospinal tracts; this differentiates ALS from a multifocal motor neuropathy.
- **Neuropsychological testing** can assist in assessment and diagnosis



Management

- No cure exists for ALS. The main focus of medical and nursing management is on interventions to maintain or improve function, well-being, and quality of life. The average survival time is **3 to 5 years** with death most commonly due to respiratory insufficiency
- **Riluzole** (Rilutek), a glutamate antagonist, has been shown to prolong survival for persons with ALS for 3 to 6 months
- **Baclofen** (Lioresal), **dantrolene sodium** (Dantrium), or **diazepam** (Valium) for muscle spasticity
- **Modafinil** (Provigil) may be used for fatigue
- Mechanical ventilation, enteral feeding



A.L.S. AMYOTROPHIC LATERAL SCLEROSIS

MEMORY TRICK

ALS = **A**dvanced **L**ife **S**upport

